

## What is Enum4linux?

Enum4linux is a **Linux-based command-line tool** used for **Windows system enumeration** using **SMB (Server Message Block)** and **NetBIOS** protocols.

## In Simple Words:

Enum4linux helps you **gather information** from a Windows machine (like usernames, shared folders, and password policies) **without logging in**, using old Windows networking features. It's often used in **penetration testing** and **CTF (Capture the Flag)** challenges to **enumerate Windows machines**.

## What Can Enum4linux Do?

### Info You Can Gather Details

|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| ✓ Usernames               | List of all user accounts on the system                       |
| ✓ Group Names             | List of groups like Administrators, Guests                    |
| ✓ Shared Folders (Shares) | Public folders over SMB, like <a href="#">\\target\public</a> |
| ✓ Operating System Info   | OS version, hostname                                          |
| ✓ Password Policies       | Min password length, lockout time, etc.                       |
| ✓ Domain & NetBIOS Info   | Workgroup/domain name, NetBIOS name                           |
| ✓ RID Brute-Forcing       | Tries common IDs to discover more users                       |

## Basic Usage

enum4linux 192.168.1.10

- This command gathers as much info as possible from the IP 192.168.1.10.

## Useful Flags

| Flag | Description                            |
|------|----------------------------------------|
| -U   | Enumerate users                        |
| -S   | Enumerate shares                       |
| -P   | Get password policy                    |
| -G   | Enumerate groups                       |
| -a   | Run <b>all</b> basic enumeration scans |

## Example: Full Scan

enum4linux -a Target IP

✅ This will try to pull:

- Usernames
- Group names
- OS details
- Shares
- Password policies

### 🔴 When Does It Work?

- Works best on **Windows systems** with:
  - **SMBv1 enabled**
  - **No firewall or open port 139/445**
  - **Weak permissions or guest access allowed**

### 🎯 Common Use Cases

| Use Case            | Why Use Enum4linux?                         |
|---------------------|---------------------------------------------|
| CTF Challenges      | To gather usernames and shares for foothold |
| Penetration Testing | Map Windows domain or network environment   |
| Red Teaming         | Get internal info for lateral movement      |

### 🔧 Behind the Scenes — Tools Used Internally

Enum4linux uses these commands in the background:

- rpcclient
- smbclient
- nmblookup
- net

### 🔒 How to Protect Against Enum4linux Attacks

| Defense                              | Description                           |
|--------------------------------------|---------------------------------------|
| Disable SMBv1 and NetBIOS            | Prevents old protocol abuse           |
| Use firewall rules                   | Block unauthorized access to port 445 |
| Disable guest accounts               | Blocks anonymous enumeration          |
| Upgrade to SMBv3 with authentication | Adds encryption and access control    |